Mertcan Demirel

Data Scientist - Credit Risk Modeling | AI & LLM



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LinkedIn | GitHub | Kaggle | Exempt

Professional Summary

- Data Scientist with 3+ years of experience, including 2.5+ years in **end-to-end credit risk model development** at Akbank: building, validating and monitoring **PD, LGD, CCF and NPL models** in line with EBA guidelines using Python and SAS. Statistics graduate ranked third in the department, with top 5% finishes in ML competitions. Alongside this work, I design and build **LLM-based systems** (multi-agent pipelines, RAG, agentic automation) and deep learning projects. **Seeking to transition into an AI / Machine Learning Engineer role** focused on LLMs and generative AI, bringing a strong statistical foundation and the rigor of regulated model development.

Work Experience

Data Scientist – Model Development | Assistant Manager, *Akbank* Mar 2024 – Present

- **Promoted** from Assistant Manager Level 4 to Level 3 in July 2025.
- Lead the end-to-end development of an EBA-compliant **Credit Conversion Factor (CCF)** model, from data preparation through validation, in full alignment with **EBA regulatory guidelines**.
- Own a module of a **Probability of Default (PD)** model end to end: data preparation, **EDA, interactive grouping, Spearman correlation** analysis and modeling.
- Develop, validate and deploy **PD and LGD models** for commercial lending portfolios using **logistic regression and machine learning** algorithms.
- Lead the development and maintenance of **Non-Performing Loan (NPL) models** within the bank's risk management framework.
- **Monitor and validate** the performance of the entire credit risk model portfolio using key metrics (Gini, PD, DR) to ensure sustained accuracy and **regulatory compliance**.
- Build **automation scripts** in Python and Excel that streamline manual data processing, improving team efficiency and reducing human error.
- Built and maintain the **KRKB AI Bulletin**, an automated twice-weekly newsletter on AI and risk management for Akbank employees: an **LLM agent** curates and summarizes news from regulatory and industry sources (EBA, Basel Committee, BDDK), with schema validation, deduplication and a GitHub Actions publishing pipeline.

Jr. Data Scientist, *Next GEO* Sep 2023 – Mar 2024

- Built and deployed **predictive regression** and **classification models** using **tree-based algorithms** to extract actionable business insights from large-scale **geospatial datasets**.

Education

Yildiz Technical University, *Bachelor's Degree - Statistics* Oct 2020 – Jun 2023
GPA: 3.67/4.00 (Ranked **Third** of the Department of Statistics)
Transferred from Ankara University, Statistics (Oct 2019 – Jul 2020, GPA: 3.93/4.00)

Skills

Programming & Databases: Python | SQL | SAS

Machine Learning & AI: Statistical Modeling | Machine Learning | Deep Learning (TensorFlow/Keras) | LLM & Agentic Workflows (Ollama, Qwen3) | RAG & Semantic Search (ChromaDB) | Speech-to-Text (OpenAI Whisper) | NLP & Sentiment Analysis | Predictive Modeling | Data Preprocessing

Tools & Platforms: FastAPI | React / Next.js | Docker | SAS Enterprise Guide | SAS Miner | SPSS | Microsoft Excel | Git / GitHub

Languages

Turkish — Native | **English** — B2

Certificates

- Data Analytics Certification - Sabancı EDU
- SAS Enterprise Guide 1: Querying and Reporting
- Python Programming for Data Science
- Artificial Intelligence Workshop
- SAS Enterprise Guide 2: Advanced Tasks and Querying
- Python A-Z™: Data Science and Machine Learning
- Python and Tensorflow for Data Science

Competitions

Türkiye İş Bankası ML Challenge #5, *Top 5%, 13th of 284* — Sep 2023 – Oct 2023

Kaggle Playground Series - Season 3, Episode 25, *Top 4%, 58th of 1632* — Nov 2023 – Dec 2023

iyzico Datathon, *Top 22%, 72nd of 328* — Jan 2024 – Feb 2024

Projects

Jarvis - Personal AI Assistant with Multi-Agent LLM Architecture — Aug 2026 – Present
Fully local AI assistant (FastAPI + React) for BIST stock analysis and training coaching. Three-agent LLM pipeline (indicator, news, orchestrator) on local Ollama/Qwen3 with RAG over historical data (ChromaDB), plus a fundamental scoring signal evaluated via rank-IC backtesting. AI training coach with Hevy API sync, deterministic progression metrics and source-cited RAG under rule-based safety constraints.

Deep Learning Projects — Jul 2026 – Present
Applied deep learning repository (Keras/TensorFlow): an MLP classifier benchmarked against logistic regression via AUC/Gini (the metrics used in credit risk), an autograd engine (micrograd) built from scratch to implement backpropagation, and a systematic comparison of optimizers, regularization and weight initialization.

Speak-to-DOCX — Dec 2025
Built a web application that transcribes audio recordings into formatted Word documents using OpenAI's Whisper model, running locally. Supports multiple audio formats, automatic language detection across 90+ languages, and timestamped transcription.

Customer Segmentation with Rule-Based and Unsupervised Learning Methods — Feb 2023 – Jul 2023
Engineered a comprehensive customer segmentation model as a graduation thesis, applying both rule-based and unsupervised learning techniques. Successfully identified and defined distinct customer personas to enable targeted business strategies.